Dual-guard repair: identical magnitude-domain data and detector

2 paths/user

4 paths/user

3–7 paths/user

RSR 5 dB Uncoded BER

RSR 12 dB Uncoded BER

RSR 20 dB Uncoded BER

10^{-1}

10^{-2}

10^{-3}

10^{-4}

10^{-5}

10^{-6}

0 20 30 40 50 60

SNR (dB)

Perfect CSI

Retrained filtered

Previous bypass

Previous repair

Dual-guard repair

Direct EM100

Fresh compact sparse

Zero observed errors (marker only)

Shading: prespecified primary window. Zero-error markers at 10^{-6} are display markers, not estimates or bounds.